COURSE OUTLINE

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HR Hiring Flow Automation with Claude

From portal filter to approved pitch, with a recruiter in control of every decision

3 days (8 hours per day) | Intermediate | Instructor-led, hands-on labs | Capstone project

About this course

Who should attend

HR operations and talent acquisition teams, HR technology and enterprise IT automation engineers, and the network and platform teams that build and host these workflows in India and worldwide.

Prerequisites

- Basic Python and the command line
- Working knowledge of your organisation's hiring process and job portal exports
- Familiarity with JSON and REST APIs
- Familiarity with an AI coding assistant is helpful, not required

What you will be able to do

- Map a real hiring flow from portal filtering to pitch and decide which steps a model may propose and which only a person may approve
- Set up Claude Code, the Claude Agent SDK and an MCP server that exposes portal exports and a candidate store as tools with strict JSON outputs
- Build a validated candidate profile pipeline that captures experience, location, salary ask and preferred working hours with unknowns marked, never guessed
- Automate call preparation, scheduling and consented recording through Twilio Programmable Voice, and score English proficiency and fit from transcripts with cited evidence
- Ship an approval-gated pitch generator delivered by email with PII minimisation, consent tracking, prompt injection defences, fairness checks, an append-only audit log and a ServiceNow onboarding handoff

Tools and technologies

Claude Code and Agent SDK | Model Context Protocol (MCP) | Python | Pydantic | FastAPI | PostgreSQL | pytest | GitHub pull requests | GitHub Actions | ServiceNow Table API | Structured JSON outputs | Twilio Programmable Voice

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MODULE 1

The hiring flow and the automation baseline

- Mapping the client flow end to end: portal filter, download, screening call, assessment, pitch, offer and onboarding handoff
- Which steps Claude may propose and which a recruiter must approve: shortlist, outreach, pitch, salary discussion
- Candidate data obligations: consent, purpose limitation and retention under India's DPDP Act and GDPR
- The teaching pattern: Claude plans and drafts, deterministic code executes, a human approves, everything is logged

HANDS-ON LAB
Stand up the lab with Docker Compose, load a synthetic portal export and draw the current versus target hiring flow with a Mermaid diagram.

DELIVERABLE
Hiring flow map with approval points marked.

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MODULE 2

Setting up Claude Code, the Agent SDK and MCP for hiring

- A Claude Code project with a CLAUDE.md that encodes hiring policy, forbidden actions and output schemas
- The Claude Agent SDK in Python: sessions, tool use, structured JSON outputs validated by Pydantic
- An MCP server exposing read-only portal export tools and a write-gated candidate store on PostgreSQL
- Secrets handling for portal, telephony and ServiceNow credentials outside prompts and repositories

HANDS-ON LAB
Build the hiring MCP server, connect it to Claude Code and have the agent list candidates through a tool call that returns validated JSON.

DELIVERABLE
Working Claude Code, Agent SDK and MCP workspace.

MODULE 3

Sourcing: filtering and downloading from the portal of choice

- Portal exports as the source of truth: CSV and XLSX downloads and sanctioned APIs, respecting each portal's terms of service
- Claude translates a job requisition into deterministic filter rules; Python applies them and reports what was excluded and why
- Deduplication across portals by email and phone with normalisation, and rejected-row reporting
- Provenance on every record: portal, export file, row number and timestamp

HANDS-ON LAB
Implement the export importer and filter engine with pytest coverage, then compare the agent's proposed filters against the recruiter's own shortlist.

DELIVERABLE
Portal import and filter pipeline with provenance.

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MODULE 4

Candidate profile schema and resume extraction

- A typed CandidateProfile: experience, skills, current and preferred location, salary ask, notice period, preferred working hours
- Claude extracts fields from resume text into the schema with a confidence per field; unknowns are marked, never guessed
- Treating resume text as untrusted input: tool-free extraction, schema validation and rejection of out-of-schema content
- Human review queue for low-confidence fields

HANDS-ON LAB
Run extraction over fifty synthetic resumes, measure field accuracy against ground truth and route the low-confidence fields to review.

DELIVERABLE
Validated candidate profile store.

DAY 2 3 modules

MODULE 5

Calling candidates: scripts, scheduling and consented recording

- Claude drafts a per-candidate call script from the requisition and the profile; the recruiter runs the call
- Deterministic scheduling of call slots by candidate time zone and preferred working hours, with reminders by email
- Twilio Programmable Voice for outbound calls, recorded consent announcements and recording retrieval
- Transcription stored against the candidate record with timestamps and the recruiter's own notes

HANDS-ON LAB
Place a call to the sandbox number, capture consent and a transcript, and attach both to the candidate record through the MCP write tool.

DELIVERABLE
Call preparation and recording workflow.

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MODULE 6

Assessing English proficiency, experience, location, salary ask and hours

- A rubric-driven scoring agent: English proficiency on CEFR-aligned bands with quoted transcript evidence for each score
- Deterministic checks for experience years, location, preferred working hours against the shift, and salary ask against the approved band
- Structured JSON score cards with confidence; low confidence and disagreement with the recruiter routed to a human
- Fairness checks: scoring only on job-relevant criteria, with protected attributes excluded from the prompt

HANDS-ON LAB
Score twenty synthetic transcripts, compare against two human raters and tune the rubric until agreement is acceptable.

DELIVERABLE
Evidence-cited candidate score card.

MODULE 7

Pitching the role: drafts within approved talking points

- Claude drafts a personalised pitch from the score card, the role and a versioned library of approved talking points
- Hard constraints enforced in code: salary band, location, working hours and benefits the company has actually approved
- Delivery by email templates only after recruiter approval; tracking replies against the candidate record
- Objection handling notes for the recruiter, never automated negotiation

HANDS-ON LAB
Generate pitches for the top ten candidates, prove the generator refuses an out-of-band salary, and route each draft through approval.

DELIVERABLE
Approval-gated pitch generator.

DAY 3 3 modules

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MODULE 8

Guardrails, approvals and audit for hiring automation

- PII minimisation with consent and retention tracking: what the model sees, what the database stores, deletion on request
- Prompt injection defences for resumes and transcripts, with tests that inject instructions and confirm they are ignored
- Approval gates with named approvers for shortlist, outreach and pitch; append-only audit log in PostgreSQL
- Bias monitoring on score distributions and a recruiter override path that is always logged

HANDS-ON LAB

Attack the pipeline with an injected resume and an unapproved pitch, watch both fail closed, and reconstruct one candidate's journey from the audit log.

DELIVERABLE

Guardrail, approval and audit layer.

MODULE 9

Orchestration, ServiceNow handoff and CI

- The pipeline as a FastAPI service with explicit state per candidate and idempotent reruns
- ServiceNow Table API records for onboarding and access requests opened on acceptance, with Flow Designer notifications
- GitHub pull requests for every prompt, rubric and schema change; GitHub Actions running pytest and evaluation sets
- Operational dashboard: time to shortlist, call completion, approval turnaround and override rate

HANDS-ON LAB

Open a pull request that changes the scoring rubric, let GitHub Actions run the evaluation set, and trigger the ServiceNow onboarding record for an accepted candidate.

DELIVERABLE

Orchestrated pipeline with CI and ServiceNow handoff.

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MODULE 10

Capstone: requisition to approved shortlist

- One job requisition in, one approved and documented shortlist out, with no candidate contacted without approval
- Acceptance review: accuracy, fairness evidence, audit completeness and a limitations statement
- Team demo and handover runbook for recruiters and IT

HANDS-ON LAB
Capstone build and demo.

DELIVERABLE
End-to-end hiring automation and runbook.

Capstone

Capstone project

A repeatable hiring pipeline that filters and downloads candidates from the portal of choice, prepares and records consented screening calls, scores English proficiency, experience, location, salary ask and preferred working hours with cited evidence, and drafts a pitch that reaches a candidate only after a recruiter approves it, with every step in an audit log.

Assessment

Assessment area	Weight	Evidence
Working functionality	25%	Demonstrable end-to-end workflow in a repeatable lab
Security and safety controls	20%	Threat model, authorization boundaries, approval gates, safe failure
Testing and evaluation	20%	Automated tests, golden dataset, measured quality and known limitations
Engineering quality	15%	Clear interfaces, error handling, reproducible setup, maintainable code
Evidence and explainability	10%	Traceable findings, sources, assumptions, logs
Documentation and presentation	10%	Architecture diagram, README, runbook, demo, lessons learned

Engineering principles used throughout

- Model output is a proposal, not a fact: every finding and generated change is validated by deterministic checks and tests.
- Authorization lives outside the model: tool access is enforced by application code, scoped credentials and policy.
- Human approval for consequential actions: deploying a mitigation, changing production configuration, rotating credentials.
- Decisions, evidence, model and tool versions, approvals and outcomes are logged without recording secrets.
- A model adapter and a tool adapter layer keep every build provider-agnostic and enterprise-portable.
- Quality is measured: false positives, false negatives, reliability, latency, cost and safe failure, not just a demo.

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